PRITHWEERAJ ACHARJEE PORAG

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Computer Science and Engineering student, AI researcher, and computational artist. I am a painter, and I believe it is high time to rethink AI art, what it is and what it costs. That question drives my research on aesthetic bias, alongside published NLP work (IEEE). I build machine-learning systems, real-time graphics, and AI products, with a focus on responsible, human-centered AI: fairness, evaluation, and interactive systems people actually use.

EDUCATION

University of Toledo, B.S. Computer Science and Engineering — Toledo, OH

In progress, 2023 to present. Expected graduation 2028.

Relevant coursework: Data Structures, Discrete Structures, Differential Equations.

RESEARCH AND PUBLICATIONS

Aesthetics Is Not a Mirror — First author, 2026

- End-to-end audit of Stable Diffusion v1.5 outputs against the LAION-Aesthetics V2 curation step.
- 200,000-trial Monte Carlo joint null test, FairFace cross-classifier replication, pre-registered held-out falsification.
- First-author poster, Midwest Graduate Symposium 2026.

Bengali Fake News Detection: A Multi-Layered LSTM Ensemble Approach — Published, IEEE QPAIN 2025

- NLP system detecting fake news in Bengali-language sources using a multi-layer LSTM ensemble.

Context-Aware Dialectal Arabic Machine Translation with Interactive Region and Register Selection — arXiv:2604.06456

- Rule-based augmentation expanding a 3,000-sentence corpus to 57,000 parallel sentences across eight regional dialects; fine-tuned mT5 for controllable region and register (second author).

DP-4DGS: Dynamic Priors for 4D Gaussian Splatting — Submitted, Midwest 2026

- Dynamic scene reconstruction with physics-informed priors for 4D Gaussian Splatting. Ongoing research with Dr. Aniruddha Ray, University of Toledo.

PROJECTS

Irris, Mobile AI Agent — irris.vercel.app

- Natural-language AI agent for the phone: give it a goal, such as clean storage, manage an app, or run a multi-step task, and Irris plans the steps and carries them out across your apps and device. (In development.)

Reiva, AI Receptionist for Clinics — reiva.haorgrix.com

- A 24/7 AI assistant for clinics: patient chat, appointment scheduling, digital prescriptions, and payment tracking in Bangla and English, with WhatsApp booking, live serial queues, reminders, and a real-time analytics dashboard.

RoboTheater, Multimodal Model Evaluation in a Live Simulation — robotheater-site.vercel.app

- Placed a frontier vision-language model (Google Gemini) inside a real-time Three.js trolley-problem simulation; a Python loop captures frames, queries the model with a zero-context prompt, and renders its decision live. Found the model reasons about the ethics correctly but fails at perception, with frequent left and right errors at low pixel scale.

EXPERIENCE

HaorGrix, Co-Founder — 2026 to present, haorgrix.com

- Co-founded and lead a digital studio delivering software, AI / MLOps, and design for global clients.

TECHNICAL SKILLS

Programming: Python, JavaScript / TypeScript, Java, SQL

Machine Learning and AI: PyTorch, scikit-learn, TensorFlow, NLP, computer vision, LLM evaluation, AI ethics, experiment design

Graphics and Web: React, Next.js, Three.js / React Three Fiber, WebGL / GLSL, Vercel

ART AND RECOGNITION

- Exhibited Artist, Mind & Soul Gallery & Studios, Toledo (Group Exhibition, Grand Opening, May 2026).
- International Recognition for Visual Art, Teravarna, 2025.
- Paintings and the book *Material: Painting in the Post-AI Era*, at material-expressions.vercel.app